Technical Report of Team MR-CAS

Authors and Institution

Ruochen Cui[1,2] **Yuhai Li**[3] **Shilong Bao**[4] **Boyu Han**[3,4] **Qianqian Xu**[4,5] **Qingming Huang**[3,4]

[1] Institute of Information Engineering, Chinese Academy of Sciences

[2] School of Cyber Security, University of Chinese Academy of Sciences

[3] School of Computer Science and Tech., University of Chinese Academy of Sciences

[4] State Key Laboratory of AI Safety, Institute of Computing Technology, CAS

[5] Beijing Academy of Artificial Intelligence

{cuiruochen25,liyuhai25}@mails.ucas.ac.cn {hanboyu23z, xuqianqian}@ict.ac.cn
{baoshilong,qmhuang}@ucas.ac.cn

Method Overview

Our method is designed for the Cooking task in the VAR 2026 AI Coach Challenge. The goal is to generate useful cooking instructions from video and provide feedback, including corrective feedback when visible mistakes occur and positive completion feedback when the step is performed correctly. An AI Cooking Coach needs to understand the full cooking process, the order of recipe steps, and the final recipe goal. It also needs to detect fine-grained action mistakes at specific moments. This is difficult for a single model to accomplish within a single reasoning pass. The model must handle global task planning, local mistake detection, timestamp localization, and natural-language feedback generation at the same time.

To address this challenge, we propose a hierarchical two-stage video understanding framework. The first stage performs Coarse-Grained Segment Generation, which builds a coarse but complete understanding of the full cooking procedure. In this stage, we use Gemini-3.5 Flash to reason over the entire video together with the recipe action list. This stage performs coarse-grained reasoning. It identifies the approximate start and end time of each recipe step and generates the corresponding instruction for each recipe step segment.

The second stage performs Fine-Grained Iterative Chunk Inspection, which conducts detailed visual inspection within each coarse segment and verifies whether the current recipe step is correctly performed. In this stage, we use GPT-5.5 for fine-grained visual reasoning within these coarse segments. For each segment, the model uses visual evidence, the current task goal, neighboring step context, and previously completed recipe actions. It then determines whether there is a visible deviation that affects the completion of the current step. If a mistake is detected, the model generates corrective feedback and assigns a timestamp when the mistake first becomes clear.

During fine-grained reasoning, we process video frames dynamically according to the segment length., which helps balance long-video coverage and local detail recognition. For short segments, we sample frames directly. For longer segments, we split them into consecutive chunks with small overlaps. For long segments with multiple chunks, we summarize what happened in previous chunks and pass the summary to later chunks. This helps the model maintain continuity when reasoning about the progress of the current recipe step. Candidate feedback from different chunks is then merged and deduplicated based on how close they are in time and semantic similarity. If no visible mistake affecting the current step is detected, the system generates a success message for the current instruction.

Finally, all instructions, feedback, and success messages are organized into the required submission format. The final output contains Instruction, Feedback, and Success messages with fine-grained timestamps.